- *Variational autoencoders* compress data into a simplified format and then recreate variations of it. This technique is used mostly for anomaly detection or creating slight modifications of existing data—usually images and video, with text less common.
- *Recurrent neural networks* are adept at handling smaller-scale sequential data like text or speech. They can play a role in text-to-speech conversion or music generation but are becoming outdated for these and other large-scale tasks as transformers become more prevalent.

1.4.2 Pretrained vs. self-hosted

You have the choice to use pretrained models offered by cloud providers or third-party services or to create and host your own LLM. While pretrained models offer ease of use, they might lack the level of customization or performance you desire. Building and hosting your own LLMs gives you maximum control over the model's training data, enabling fine-tuning to achieve optimal performance for your specific application. However, this path requires significant computational resources and expertise in machine learning.

For our implementation, we'll use pretrained models via public APIs such as Gemini and OpenAI, which offer a practical way to access state-of-the-art language models without the complexity of hosting and maintaining them ourselves.

1.4.3 Performance considerations

It's important to evaluate a model's latency (inference time) and resource requirements to ensure it delivers a smooth user experience without overloading your infrastructure—or your users' costs. That means defining your user interaction and data flow and finding a model that works well for this flow. In particular, think about

- *UI design*—Determine how users will provide input to the model (e.g., text prompts, selection options) and how the generated output will be presented in the UI.
- *Data preprocessing*—Analyze how user input might need to be transformed or formatted before being fed to the chosen AI model.
- *Output postprocessing*—Consider any postprocessing steps needed on the model's output before integrating it into the final user experience.

1.5 Generative vs. traditional AI

AI is the field of computer science concerned with creating machines that solve problems by mimicking human cognitive abilities. *Machine learning* and *deep learning* are types of AI that enable computers to learn patterns from labeled data and improve their performance without being explicitly programmed. Instead of writing a series of instructions for a computer to identify a picture of a cat, we train a machine learning model to recognize images of cats. Then we can show the computer a picture of a dog, and it will respond: "No, that is not a cat." Figure 1.5 shows the process of training a model to recognize cats.